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CS2024 - Theory of Computation

The beginning

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1.

Automaton

Here we go again

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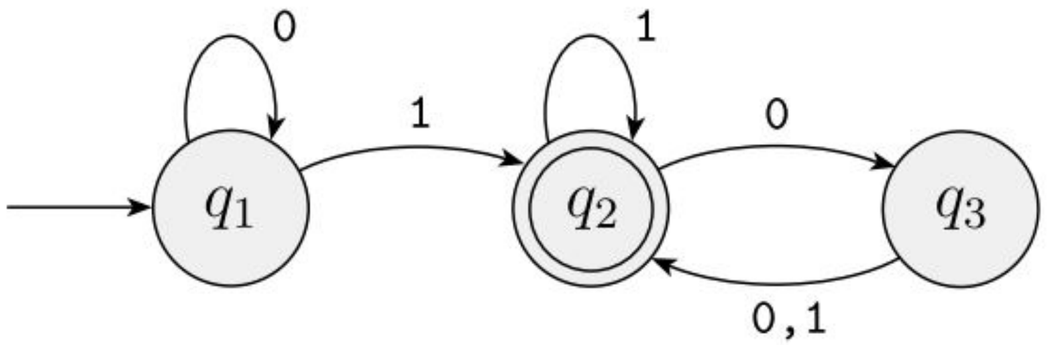
Automata *Theory*

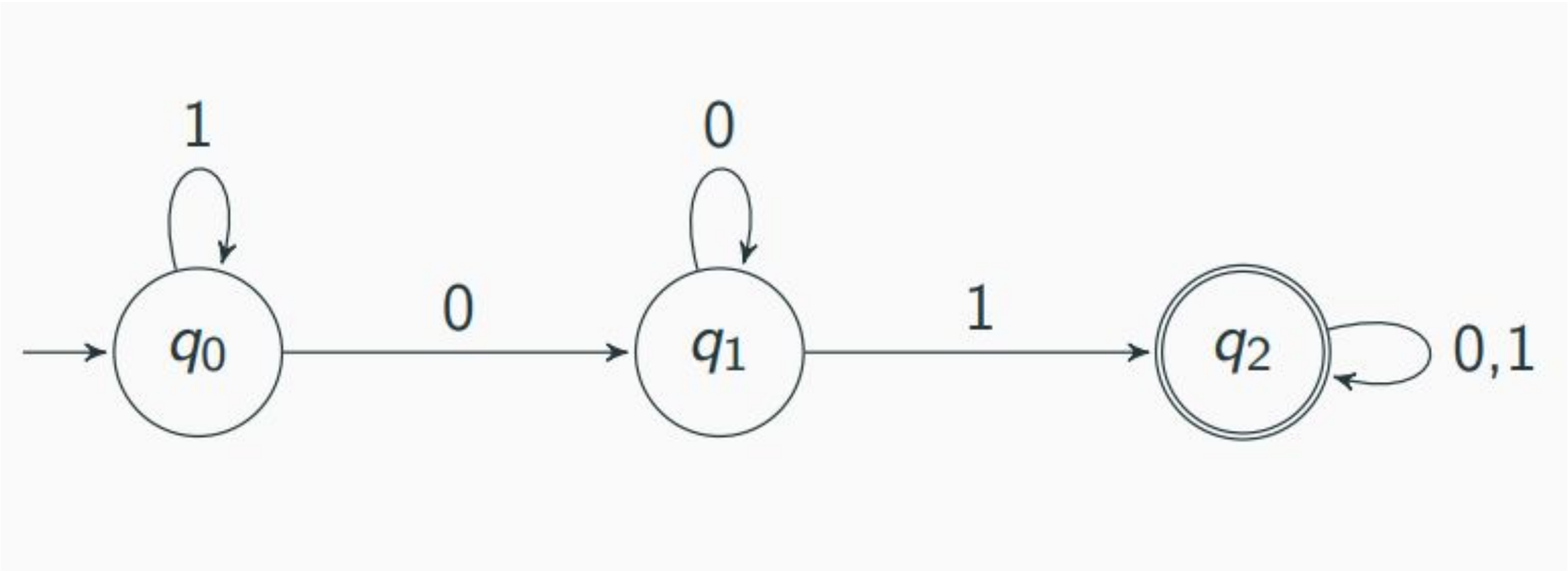
- **What is a computer?**
  - Complicated question
- We need to formalize the definition of computer
- First formalized an ideal version: **computational model**
  - **First model:**
    - Finite state machine, or **finite automaton**

A (Deterministic Finite) *automaton* M_1



- **State Diagram:** Figure
- **States:** q_1, q_2, q_3
- **Start state:** q_1
- **Accept state:** q_2
- **Transitions:** the arrows

A (Deterministic Finite) *automaton* M_1



DFA

DEFINITION 1.5

A ***finite automaton*** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

1. Q is a finite set called the ***states***,
2. Σ is a finite set called the ***alphabet***,
3. $\delta\colon Q \times \Sigma \longrightarrow Q$ is the ***transition function***,[1]
4. $q_0 \in Q$ is the ***start state***, and
5. $F \subseteq Q$ is the ***set of accept states***.[2]

Formal Definitions

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a finite automaton and let $w = w_1 w_2 \cdots w_n$ be a string where each w_i is a member of the alphabet Σ. Then M ***accepts*** w if a sequence of states $r_0, r_1, \ldots, r_n$ in Q exists with three conditions:

1. $r_0 = q_0$,
2. $\delta(r_i, w_{i+1}) = r_{i+1}$, for $i = 0, \ldots, n-1$, and
3. $r_n \in F$.

Automata *Theory*

- M **recognizes** a language L if

  L = {ω| M accepts ω}

- We also write **L = L(M)**

- A language L is called a **regular language** if there exists some finite Finite automaton M that recognizes it

Examples

- El lenguaje de todas las cadenas de 0's y 1's que consisten en n 0's seguidos de n 1's, para algún n ≥ 0
  - **{ε, 01, 0011, 000111, . . .}**
- El lenguaje de todas las cadenas de 0's y 1's que tienen la misma cantidad de 0's y 1's
  - **{ε, 01, 10, 0011, 0101, 1001, . . .}**
- El lenguaje de todas las cadenas de 0's y 1's que representan números primos
  - **{10, 11, 101, 1011, . . .}**

Class Activity

Diseñe autómatas finitos para los siguientes lenguajes. El alfabeto es siempre {0, 1}

a. Cadenas con un número impar de 1's
b. Cadenas con un número par de 1's
c. Cadenas con un número de unos múltiplo de 3
d. Cadenas que tienen un número par de 0s siguiendo al último 1, y que tienen al menos un 1
e. Cadenas que tienen al 001 como subcadena
f. Cadenas que tienen al 011 como subcadena
g. Cadenas con tres ceros consecutivos
h. Cadenas cuyo segundo dígito empezando desde la derecha es 1

Class Activity

Diseñe autómatas finitos para los siguientes lenguajes. El alfabeto es siempre {0, 1}

a. Cadenas que comienzan en 1 y terminan en 0
b. Cadenas que tienen al 0101 como subcadena
c. Todas las cadenas excepto la cadena vacía
d. Cadenas cuyo tercer dígito empezando desde la derecha es 1
e. {ε, 0}
f. El conjunto vacío
g. Cadenas cuya longitud es máximo 5
h. Cadenas con un número impar de 1's y un número impar de 0's

Regular Operations

- Let A and B two languages
- Regular operations
  - **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$
  - **Concatenation:** $A \cdot B = \{xy : x \in A , y \in B\}$
  - **Estrella:** $A^* = \{x_1x_2 \ldots x_k : x_i \in A, \text{ and } k \geq 0\}$

Regular Operations

- A = {good, bad} and B = {boy, girl}
  - A ∪ B = {good, bad, boy, girl}
  - A · B = {goodboy, goodgirl, badboy, badgirl}
  - A^* = {ε, good, bad, goodgood, goodbad, badgood, badbad, goodgoodgood, goodgoodbad, . . .}

Regular Operations

- Let REGULAR, the set of regular languages L
  - Is REGULAR closed under the **union** operation?
  - Is REGULAR closed under **concatenation**?
  - Is REGULAR closed under **star** operation?

Theorem

REGULAR is closed under the union operation.

If L_1 and L_2 are two regular languages, then $L_1 \cup L_2$ is also a regular language

Proof Idea

- Proof by construction
- Since L_1 is regular, there exists a FDA A_1 that recognizes L_1
- Since L_2 is regular, there exists a FDA A_2 that recognizes L_2
- We will build A using A_1 and A_2

Proof Idea

- Let $L = L_1 \cup L_2$
- Let $A_1 = \{Q_1, \Sigma, \delta_1, q_1, F_1\}$
- Let $A_2 = \{Q_2, \Sigma, \delta_2, q_2, F_2\}$
- We define $A = \{Q, \Sigma, \delta, q, F\}$ as
- $Q = Q_1 \times Q_2$
- $\delta((r_1,r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$
- $q_0 = (q_1, q_2)$
- $F = \{(r_1,r_2) : r_1 \in F_1 \text{ or } r_2 \in F_2)\}$

Theorem

REGULAR is closed under concatenations.

If L_1 and L_2 are two regular languages, then $L_1 \cdot L_2$ is also a regular language

Proof Idea

- Spooky
- How can a FDA know the right partition to the string such that the first half is accepted by A1 and the second by A2?
- We need a more permissive model
- We need **NONDETERMINISM**

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Thanks

Let’s keep finding mistakes

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